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"When Algorithms Go Rogue: Whistle-blower's and the Hidden Risks of AI-powered AML in Algorithmic Trading?"

Abstract

This study delves into the intersection of AI and algorithmic trading, examining how these technologies can be manipulated to gain unfair advantages in financial markets, particularly in facilitating insider trading. The inherent complexity of AI algorithms often makes it challenging to detect such illicit activities using traditional monitoring methods. This complexity poses significant challenges to existing deterrence strategies, which predominantly focus on economic and legal repercussions. The study proposes solutions for designing and implementing AI-powered AML systems that are more transparent, accountable, and effective. These improvements aim to enhance detection capabilities and foster a culture of whistleblower protection within algorithmic trading firms. Through these measures, the financial industry can better safeguard against the exploitation of AI in trading and ensure a more secure and fair market environment.

CHAPTER -1 Introduction

The financial sector, a realm traditionally defined by meticulous human analysis and intuition, is witnessing a transformative shift. Artificial intelligence (AI) is rapidly weaving its way into the fabric of finance, breathing life into sophisticated algorithms that are revolutionizing the way we trade. Algorithmic trading, the practice of using computer programs to execute trades based on predefined rules, has long been a fixture in financial markets. However, the integration of AI into these algorithms marks a paradigm shift, unlocking a new era of efficiency, automation, and potentially, unforeseen risks. The allure of AI-powered algorithmic trading lies in its ability to process vast quantities of data at lightning speed. Unlike human traders, who are limited by cognitive constraints, AI algorithms can analyse mountains of financial information, including historical price data, news feeds, social media sentiment, and even satellite imagery, to identify intricate patterns and potential trading opportunities. This prowess translates into faster and more precise execution of trades, minimizing the impact of human emotions and reaction times. The automation of trading strategies further bolsters efficiency by freeing up human traders to focus on higher-level tasks like portfolio management and risk assessment. The benefits of these technological advancements extend beyond mere speed and efficiency. AI algorithms, particularly those powered by machine learning, possess the remarkable ability to continuously learn and adapt. By analysing past successes and failures, these algorithms can refine their trading strategies, leading to potentially superior performance over time. Additionally, AI can be instrumental in uncovering hidden correlations within complex datasets, allowing for the discovery of previously unseen trading opportunities. This not only enhances potential returns but also contributes to a more dynamic and efficient financial market. However, the very features that render AI-powered algorithmic trading so powerful also raise critical concerns. The opacity of these algorithms, often shrouded in layers of complex code, can create a situation where regulators and even the users themselves struggle to fully understand their inner workings. This lack of transparency can create fertile ground for unintended consequences, potentially leading to algorithmic biases that distort market behaviour. Perhaps the most concerning aspect of this algorithmic evolution is the potential for exploitation. The speed and anonymity afforded by AI-powered trading could create new avenues for criminal activities like insider trading and money laundering. Malicious actors could exploit the intricate decision-making processes of these algorithms to inject false information or manipulate market sentiment, thereby gaining an unfair advantage.

The spectre of algorithmic manipulation further exacerbates the existing concerns surrounding market volatility. High-frequency trading (HFT) algorithms, a subset of AI-powered trading strategies, are notorious for their lightning-fast execution of trades, often exploiting minuscule price discrepancies. While HFT can contribute to market liquidity, the sheer volume and speed of these trades can amplify market volatility, potentially leading to flash crashes where prices plummet dramatically in a short period. Additionally, the interconnected nature of these algorithms raises the risk of cascading failures. If one algorithm malfunctions or encounters unexpected market conditions, it could trigger a domino effect, disrupting the entire financial system. The rapid evolution of AI-powered algorithmic trading necessitates a multifaceted approach to ensure its responsible and sustainable integration into the financial landscape. Regulatory bodies must strive to strike a balance between encouraging innovation and safeguarding market integrity. This may involve mandating greater transparency in algorithmic design, requiring robust back testing procedures, and establishing clear guidelines for the use of AI in financial markets. Additionally, fostering a culture of responsible development within the financial technology (FinTech) industry is crucial. Developers must prioritize ethical considerations throughout the design process, ensuring that their algorithms are not susceptible to manipulation or exploitation.

Moreover, ongoing research and development are essential to mitigate the risks associated with AI-powered algorithmic trading. Exploring techniques to enhance transparency, such as explainable AI (XAI) frameworks, could allow regulators and users to better understand how these algorithms arrive at their decisions. Additionally, developing more robust risk management frameworks specific to AI-powered trading will be vital in preventing algorithmic malfunctions from cascading into systemic crises. The realm of AI-powered algorithmic trading is a dynamic landscape, teeming with a diverse array of approaches. Each strategy leverages the unique capabilities of AI to extract insights from vast datasets and identify profitable trading opportunities. This section delves into the three prominent approaches – machine learning, deep learning, and statistical arbitrage – exploring their strengths, limitations, and potential implications. Machine learning (ML) sits at the heart of many AI-powered trading strategies. ML algorithms are designed to learn from historical data and continuously improve their performance over time. This "learning" process involves training the algorithm on a vast dataset of market data, including historical prices, trading volumes, economic indicators, and even social media sentiment. By analyzing these intricate

relationships, the algorithm identifies patterns and correlations that can predict future market movements.

One popular ML technique in algorithmic trading is supervised learning. In this approach, the algorithm is presented with labeled data, where each data point has a corresponding outcome. For example, the data might include historical price data and a label indicating whether the price went up or down the following day. By analyzing this labeled data, the ML algorithm learns to recognize patterns that can be used to predict future price movements. Another prevalent technique is unsupervised learning. Here, the algorithm is tasked with uncovering hidden patterns in unlabeled data. This approach is particularly useful for identifying previously unseen relationships within complex datasets. For instance, an unsupervised learning algorithm might analyze historical price data and social media sentiment to discover how social media buzz affects specific stock prices. The adaptability of ML algorithms is a key advantage. As they are exposed to more data and market fluctuations, they can refine their predictive models, potentially leading to superior performance over time. Additionally, ML algorithms can be customized based on specific trading strategies. A risk-averse investor might prioritize algorithms that focus on identifying low-risk, high-probability trades, while a more aggressive trader could utilize strategies that aim to capitalize on short-term market movements. However, the effectiveness of ML is heavily reliant on the quality and quantity of data used for training. Poor-quality data can lead to biased algorithms that generate unreliable predictions. Additionally, ML models can struggle to adapt to sudden market shifts or unforeseen events, potentially resulting in significant losses.

Deep learning (DL), a subfield of ML, takes the concept of learning algorithms to a whole new level. Deep learning algorithms are inspired by the structure and function of the human brain, utilizing artificial neural networks (ANNs) with multiple layers. These layered networks allow DL algorithms to process information in a hierarchical manner, extracting increasingly complex features from the data. In the context of algorithmic trading, DL algorithms can analyze vast amounts of unstructured data like news articles, social media feeds, and satellite imagery. By processing this data through the layers of the ANN, the algorithm can identify subtle nuances and hidden relationships that might escape traditional ML approaches. This ability to glean insights from diverse data sources allows DL algorithms to make more

informed and potentially more accurate predictions. For example, a DL algorithm might analyze news articles and social media sentiment to gauge investor confidence towards specific companies. This information, combined with historical price data, could be used to predict market trends and identify potential buying or selling opportunities. Additionally, DL algorithms can be particularly adept at recognizing complex patterns in technical indicators, which are graphical representations of historical price and volume data used by some traders to predict future price movements.

The traditional methods employed by AML systems to detect suspicious activity are being challenged by the rise of AI and algorithmic trading. These systems typically rely on rule-based algorithms and keyword filters to identify red flags associated with money laundering. However, the complex and dynamic nature of AI-driven transactions can easily bypass these filters. For example, AI could be used to structure transactions into smaller amounts to avoid triggering AML reporting thresholds. Additionally, AI could be employed to generate synthetic data that mimics legitimate transactions, further obfuscating the true nature of the underlying activity. These advancements require AML systems to become more sophisticated and adapt to the evolving techniques of financial criminals.

Combating financial crime in the age of AI requires a multi-pronged approach. Regulatory bodies need to adapt and update existing AML regulations to account for the specific risks posed by AI and algorithmic trading. This may involve implementing stricter reporting requirements for high frequency trading activity and mandating the use of explainable AI models in financial applications. Furthermore, collaboration between financial institutions and regulatory bodies is crucial. Sharing data and developing advanced analytics tools specifically designed to detect AI-driven financial crime can be instrumental. Additionally, investing in research and development of AI-powered AML systems can help identify and disrupt criminal activity before it occurs. The integration of AI into finance raises important ethical questions. As algorithms become more sophisticated, the line between legitimate AI-driven trading and manipulative or exploitative practices can become increasingly blurred. Financial institutions need to establish clear ethical guidelines for the development and deployment of AI in trading activities. Furthermore, there's a need for increased transparency and accountability in the development of AI algorithms used in finance. This can help mitigate the risk of bias or unintended consequences that could inadvertently facilitate financial crime. Anti-

Money Laundering (AML) regulations are crucial for safeguarding the financial system from criminal activity. Whistleblowers play a vital role in exposing money laundering schemes, yet they often face significant challenges that can deter them from coming forward. This essay explores the specific hurdles these brave individuals encounter and proposes potential solutions to empower them. One of the primary challenges faced by AML whistleblowers is the fear of retaliation. The individuals or institutions involved in money laundering often wield considerable power and influence. Whistleblowers may fear losing their jobs, being ostracized by colleagues, or even facing legal repercussions for disrupting the status quo. This fear can create a chilling effect, silencing those with valuable information.

Financial institutions have intricated internal controls designed to detect and report suspicious activity. However, whistleblowers may not fully understand these procedures, making it difficult for them to identify and report red flags effectively. Additionally, some institutions may have a culture of silence, discouraging employees from questioning suspicious activity. Furthermore, the legal framework for protecting whistleblowers in AML cases can be inadequate. Current whistleblower protection laws might not offer sufficient safeguards against retaliation, particularly for those working in the private sector. Additionally, the process of reporting AML violations can be lengthy and cumbersome, discouraging whistleblowers from pursuing official channels. To empower AML whistleblowers and strengthen the fight against money laundering, several solutions can be implemented. Firstly, robust legal protections are essential. Strengthening whistleblower protection laws and ensuring their effective implementation will provide a safety net for those who come forward. This includes safeguarding whistleblowers from retaliation and ensuring their anonymity when necessary. Secondly, fostering a culture of open communication within financial institutions is crucial. Institutions should actively encourage employees to report suspicious activity and provide clear channels for doing so. Training programs can educate employees on AML procedures and empower them to identify red flags. Finally, creating a centralized, independent body specifically for receiving and investigating AML whistleblower reports can be highly beneficial. This independent body would ensure a safe and confidential reporting process while expediting investigations.

The purpose of this study is to determine how algorithmic trading and artificial intelligence (AI) can be used to launder money and what the shortcomings are in the present AI-powered

AML systems' ability to identify such actions. Difficulties that algorithmic trading firms' whistleblowers encounter when disclosing problems with AI-powered AML systems. What distinguishes these issues from conventional insider trading situations. The design and implementation of AI-powered AML systems should priorities transparency, accountability, and effectiveness in reducing the risks associated with algorithmic trading, all the while promoting a culture that protects whistleblowers. Does this act in that way?

RESEARCH QUESTION

1-How can AI and algorithmic trading be exploited for money laundering activities, and what are the limitations of current AI-powered AML systems in detecting such activities?

2-What specific challenges do whistle-blowers face when reporting issues with AI-powered AML systems in algorithmic trading firms? How do these challenges differ from traditional insider trading scenarios?

3- How can AI-powered AML systems be designed and implemented to be more transparent, accountable, and effective in mitigating the risks associated with algorithmic trading, while fostering a culture of whistle-blower protection?

CHAPTER -2 Literature Review

Trading procedures have undergone substantial changes because of artificial intelligence's (AI) quick development and incorporation into financial markets. Thanks to AI-powered algorithmic trading, which allows for sophisticated and high-frequency trading tactics, market dynamics have completely changed. This development has sparked worries about the possibility of abusing the technology to obtain unfair advantages, including making insider trading easier, though. Furthermore, traditional approaches to identifying and discouraging such criminal activity face significant problems due to the complex nature of AI systems. With these advancements, the need of whistleblowers in exposing questionable activity grows, even though they face difficulties in AI-driven settings. This literature review explores the intersections of AI in algorithmic trading, the limitations of current deterrence methods, the blind spots in AI-powered anti-money laundering (AML) systems, and the specific challenges faced by whistleblowers in this context. It also proposes potential improvements for AI-powered AML systems to foster a more ethical algorithmic future.

The use of sophisticated algorithms to make trading choices at frequencies and speeds faster than those of a human has led to the rise of algorithmic trading as a major player in the financial markets. Aldridge and Krawciw (2017) state that because algorithmic trading can analyze large volumes of data quickly and execute trades quickly, it accounts for a sizable share of trading activity in key markets. But this technological advantage can also be utilized for insider trading, which is the unfair use of material, non-public knowledge to one's benefit in trading. The literature highlights that the opacity and complexity of AI algorithms can obscure illicit activities. As noted by Cao et al. (2021), AI algorithms can be designed to detect and act on subtle patterns in market data that might indicate the presence of non-public information, thus facilitating insider trading. Traditional surveillance systems, which rely on identifying unusual trading patterns and direct connections to insider information, may struggle to detect such sophisticated schemes. Moreover, Leins (2020) argues that the "black box" nature of AI, where the decision-making processes of algorithms are not transparent even to their developers, exacerbates the difficulty of detecting insider trading. Regulators and compliance officers face difficulties since they must discover and prosecute market abuses based on evidentiary that is both clear and understandable. Therefore, the intricacy of AI algorithms can be used as a

weapon for market efficiency as well as a cover for illegal activity, increasing the demand for sophisticated detection techniques. AI has the potential to be used in insider trading because of its quick processing speed and subtle action on confidential information. Fernandes and Ferreira (2016) claim that artificial intelligence (AI) algorithms may be used to recognize and take advantage of information asymmetries, which will facilitate traders with privileged information to execute trades without raising red flags. The danger of insider trading is increased by AI's capacity to combine and evaluate information from various sources, giving traders better ability to make decisions based on confidential knowledge. Regulators have a difficult time reducing the hazards brought on by insider trading powered by AI. According to Al-Jarrah et al. (2019), the advanced application of AI and ML in trading may be beyond the capabilities of current legal frameworks. Artificial intelligence (AI)-driven trading techniques require sophisticated regulatory tools and monitoring systems to comprehend and analyses them. In addition, ethical issues need to be considered to guarantee that AI is applied in a way that advances market transparency and justice. Many technical remedies have been put up to counter the possibility of AI-facilitated insider trading. According to Feng et al. (2021), the development of advanced AI-driven surveillance systems that can detect subtle and complex trading patterns is crucial. These systems would leverage the same AI and ML technologies to monitor trading activities and identify suspicious behaviors that traditional methods might miss. Additionally, fostering collaboration between regulators, financial institutions, and technology developers is essential to create a comprehensive approach to managing the risks associated with AI in trading (Ahern, 2017) (Anita Anand b, 2019).

Whistleblowing has traditionally played a crucial role in exposing financial misconduct. However, the rise of AI in trading introduces new limitations to current deterrence methods, which are primarily focused on economic and legal repercussions. As described by Bressler and Bressler (2019), whistleblowers face substantial risks, including retaliation, career damage, and legal consequences. These risks are compounded in AI-driven environments where the nature of evidence is more complex and harder to interpret. The importance of whistleblowers in uncovering suspicious activity cannot be overstated. According to Dyck, Morse, and Zingales (2010), a significant proportion of corporate fraud is exposed by employees who blow the whistle, highlighting the critical role they play in maintaining market integrity. However, in the context of AI-powered crimes, traditional whistleblower protections and incentives may be insufficient. Challenges specific to whistleblowing in AI-driven environments include the difficulty in obtaining concrete evidence due to the algorithmic complexity and the fear of

retaliation in a highly technical and potentially opaque organizational culture. Banerjee et al. (2022) emphasizes the need for a supportive environment that encourages whistleblowers to come forward without fear of retribution. This environment is particularly crucial in tech-driven sectors where understanding and interpreting AI algorithms require specialized knowledge. Whistleblowers are vital to exposing AI-related wrongdoing in this intricate situation. Whistleblowers offer a human aspect that can recognize and analyze suspicious activity within an organizational environment, in contrast to automated monitoring methods. They frequently possess insider information, and a sophisticated grasp of business processes that outside auditors or regulators might overlook. This insider knowledge is extremely helpful in identifying subtle and complex manipulations made possible by artificial intelligence systems. Authorities can be informed by whistleblowers about unethical behavior, data manipulation, biassed algorithms, and other AI-driven wrongdoing that may not be immediately obvious. Research on AI crimes can get greater insights into the internal processes that facilitate these crimes and how they are perpetrated by include the perspective of whistleblowers.

The integration of AI in trading creates new blind spots in identifying and reporting suspicious activities. Whistleblowers in algorithmic trading firms may struggle to recognize malfeasance due to the sophisticated nature of AI systems. As argued by Singh et al. (2018), AI algorithms can mask illicit activities through intricate trading patterns that appear legitimate on the surface. This complexity makes it difficult for individuals without deep technical expertise to detect wrongdoing. Current AI-powered AML systems also face significant limitations in detecting money laundering schemes that leverage algorithmic trading. According to a study by Goldstein, Jiang, and Karolyi (2019), traditional AML systems are designed to detect suspicious activities based on predefined rules and patterns. However, AI-driven trading can generate new, unpredictable patterns that evade these systems. The adaptability and learning capabilities of AI make it possible for money launderers to continuously modify their strategies to stay ahead of detection mechanisms. Moreover, Fuster et al. (2018) highlight that AI algorithms can be trained to exploit regulatory loopholes and create complex transaction networks that are difficult to trace. This adaptability challenges existing AML frameworks that rely on static rules and historical data. Consequently, there is a pressing need for more dynamic and robust AML systems capable of adapting to the evolving tactics of financial criminals. Systems that prevent money laundering are crucial in the battle against financial crime. These technologies identify and flag potentially suspicious transactions using artificial intelligence.

Algorithmic trading-based money laundering schemes, however, pose significant difficulties that current AML systems are ill-prepared to manage. The ability of money launderers to take advantage of the speed and intricacy of algorithmic trading is one of the main drawbacks. As Van Vlasselaer et al. (2017) highlight, AI-powered AML systems typically rely on pattern recognition and anomaly detection. Nevertheless, complex money laundering techniques can be made to look like real trading activity, making it challenging to tell them apart. Through the distribution of unlawful monies over multiple transactions and trading platforms, money launderers can efficiently avoid being discovered. Moreover, Gai et al. (2016) noted that the integration and analysis of data from various sources may present challenges for current AML systems. A complicated web of data is produced by algorithmic trading, which frequently incorporates several financial instruments and cross-border transactions. Sophisticated money launderers can take advantage of these loopholes in AML systems' limited capacity to gather and evaluate this data in real-time. The versatility of money laundering methods is a serious concern. Bala et al. (2019) claim that to keep ahead of detection technologies, money launderers constantly adapt their techniques. Though sophisticated, AI-powered AML systems might not react quickly enough to emerging money laundering tactics. Many rule-based systems are static, which makes them susceptible to the creative and dynamic strategies used by money launderers in the context of algorithmic trading. Many rule-based systems are static, which makes them susceptible to the creative and dynamic strategies used by money launderers in the context of algorithmic trading. Additionally, augmenting anomaly detection with AI and network analysis can improve it. Network analysis can uncover links and transactions that seem innocent in isolation but are suspicious when seen as part of a larger network, as noted by Gao and Ye (2007). Artificial intelligence (AI)-powered AML systems can find hidden links and patterns suggestive of money laundering by mapping and evaluating transaction networks. Real-time data analysis integration is another crucial topic. Conventional AML systems frequently rely on cyclical batch processing, which can cause a lag in the identification of questionable activity. Advanced AI technologies offer real-time analytics to deliver prompt insights and alarms, facilitating prompt intervention and mitigation. Real-time data processing is crucial in the quick-paced world of algorithmic trading, where delays might result in lost opportunities to identify and stop financial crimes, as Chen et al. (2018) point out.

The effectiveness of AML compliance cannot rely solely on the deterrence effect of regulations. While stringent regulations and harsh penalties are essential, they must be complemented by advanced technological solutions that address the new risks introduced by

algorithmic trading. As suggested by Wechsler (2020), focusing solely on deterrence overlooks the proactive measures needed to prevent and detect illicit activities in real-time. Improving AI-powered AML systems involves enhancing the transparency of AI algorithms used in trading. Explainable AI (XAI) techniques, which aim to make AI decision-making processes more interpretable, can play a crucial role in this regard. According to Arrieta et al. (2020), XAI can provide human oversight by making it easier to understand how algorithms arrive at their decisions. This transparency can help regulators and compliance officers identify and investigate suspicious activities more effectively. Additionally, implementing better anomaly detection mechanisms is vital. As noted by Goodell et al. (2021), traditional anomaly detection systems may not be sufficient in the context of AI-powered trading due to the complexity and high frequency of transactions. Advanced AI techniques, such as deep learning and reinforcement learning, can be leveraged to develop more sophisticated anomaly detection models that can identify subtle and complex patterns indicative of money laundering and other illicit activities. The creation of explainable AI (XAI) models, which offer comprehensible and comprehensible insights into the decision-making process involved in trading, is one strategy for increasing transparency. Doshi-Velez and Kim (2017) claim that XAI can make it simpler to identify and investigate anomalies by bridging the gap between intricate algorithmic processes and human comprehension. Financial firms can guarantee that their trading algorithms are not only efficient but also transparent and comprehensible to regulatory agencies by putting XAI into practice. Furthermore, it might be necessary to mandate the disclosure of important algorithmic parameters and trading tactics. Even if this creates issues with intellectual property and competitive advantage, a sensible strategy that shares important information with regulators under stringent confidentiality agreements could be used. According to Zingales (2017), this would improve regulatory supervision without jeopardizing trading algorithms' unique character. An essential component of efficient AML systems is anomaly detection. Conventional techniques depend on pre-established guidelines and patterns, which may not be enough to detect complex money laundering operations. Although AI-powered systems can detect anomalies more dynamically and adaptively, they require ongoing development to stay up to date with changing security risks. Using machine learning algorithms, which can recognize intricate and non-linear patterns suggestive of money laundering, is one promising strategy. Ngai et al. (2011) claim that unsupervised learning techniques—like neural networks and clustering—can identify abnormalities without depending on pre-established rules. These methods can analyze enormous volumes of data and spotting minor, previously undetected suspicious activity patterns.

Whistleblowers face unique challenges when reporting issues with AI-powered AML systems in algorithmic trading firms. These challenges differ significantly from traditional insider trading scenarios. One major challenge is the difficulty in obtaining evidence due to the algorithmic complexity. As pointed out by Kroll et al. (2021), understanding and interpreting the outputs of AI algorithms require specialized knowledge that whistleblowers may not possess. This lack of understanding can hinder their ability to provide clear and compelling evidence of wrongdoing. Fear of retaliation is another significant concern. In a tech-driven environment, whistleblowers may face more severe and sophisticated forms of retaliation. As discussed by Rehg et al. (2008), the fear of losing one's job, facing legal action, or being ostracized by peers can deter potential whistleblowers from coming forward. This fear is exacerbated in AI-driven environments where the organizational culture may prioritize technological advancements and profit over ethical considerations. Furthermore, the rapid pace of technological change in algorithmic trading firms can create additional challenges. Whistleblowers may struggle to keep up with the latest developments and identify when and how AI algorithms are being misused. As noted by Lewis and Uys (2021), the dynamic nature of AI technologies requires continuous learning and adaptation, which can be a daunting task for potential whistleblowers.

Designing and implementing AI-powered AML systems that are more transparent, accountable, and effective is crucial for fostering a more ethical algorithmic future. Explainable AI techniques can enhance transparency by providing insights into how AI algorithms make decisions. This transparency can facilitate human oversight and improve the detection of suspicious activities. As stated by Ribeiro, Singh, and Guestrin (2016), XAI can help bridge the gap between the technical complexity of AI and the need for understandable and actionable insights. Accountability is another key aspect. Clear lines of responsibility should be established in case of malfunctions or misuse of AI algorithms. As suggested by Gasser and Almeida (2017), accountability frameworks should ensure that individuals and organizations can be held responsible for the outcomes of AI systems. This accountability can deter misconduct and promote ethical behavior in algorithmic trading firms. Moreover, improving the effectiveness of AI-powered AML systems requires continuous innovation in detection mechanisms. Advanced machine learning techniques, such as neural networks and reinforcement learning, can enhance the ability to detect complex and evolving patterns of

suspicious activity. According to Ngai et al. (2011), these techniques can improve the accuracy and reliability of AML systems, making it harder for financial criminals to evade detection.

Fostering a culture of whistleblower protection within algorithmic trading firms is also essential. As highlighted by Near and Miceli (1996), organizations should create an environment that encourages employees to report suspicious activities without fear of retaliation. This can be achieved through robust whistleblower protection policies, training programs, and a strong ethical culture that values transparency and integrity. The integration of AI in algorithmic trading presents both opportunities and challenges. While AI can enhance market efficiency and trading strategies, it also introduces new risks for insider trading and money laundering. Traditional detection and deterrence methods may be insufficient in addressing these risks due to the complexity and opacity of AI algorithms. Whistleblowers play a critical role in uncovering suspicious activities, but they face unique challenges in AI-driven environments. To create a more ethical algorithmic future, it is essential to design AI-powered AML systems that are transparent, accountable, and effective. Enhancing transparency through explainable AI, establishing clear accountability frameworks, and improving detection mechanisms can help address the new risks posed by algorithmic trading. Additionally, fostering a culture of whistleblower protection is crucial for encouraging the reporting of suspicious activities and maintaining market integrity.

CHAPTER -3 Methodologies

The study examines insider trading within multinational and domestic firms over an extensive period from 1987 to 2019, utilizing a comprehensive dataset that includes insider trade data from Thomson Reuters insider Form 4 filings, stock return data from the Center for Research in Security Prices (CRSP), and financial statement data from Compustat. To refine the sample and mitigate the influence of extreme outliers, firms with a market capitalization below $25 million are excluded, and control variables are winsorized at the 99% and 1% levels. The identification of insider trading is meticulously structured around the concept of insider trade months, defined at the firm-month level. Specifically, purchase months are those where all insiders at a firm are net purchasers of stock, while sale months are characterized by all insiders at a firm being net sellers. This dichotomy allows for a clear distinction between different types of insider activities. The study further differentiates firms into multinational and domestic categories based on the presence of foreign sales, with those having such sales labeled as multinational. This classification is fundamental to the subsequent analysis, enabling a focused investigation into the differences in insider trading behaviors between these two groups.

A significant methodological decision in the study is the exclusion of routine insiders, identified using the methodology of Cohen et al. (2012), to hone in on informed insider trading. This exclusion is pivotal as it aims to isolate trades made based on unique, non-routine information that insiders may possess. The summary statistics of the sample reveal a substantial dataset, encompassing 67,248 purchase months and 140,306 sale months. Notably, multinational firms represent a significant portion of this dataset, with 31,721 multinational purchase months and 82,412 multinational sale months, compared to 35,527 domestic purchase months and 57,894 domestic sale months. These figures underline the prominence of multinational firms in the insider trading landscape. An intriguing finding from the summary statistics is that multinational firms tend to be larger in size than domestic firms but exhibit slightly lower book-to-market ratios. This observation sets the stage for further analysis into the implications of firm size and valuation on insider trading. The standard deviation of returns, serving as a measure of return volatility, shows only small differences between multinational and domestic firms, suggesting that volatility may not be a major differentiating factor in the context of insider trading. (refer Appendix for Table)

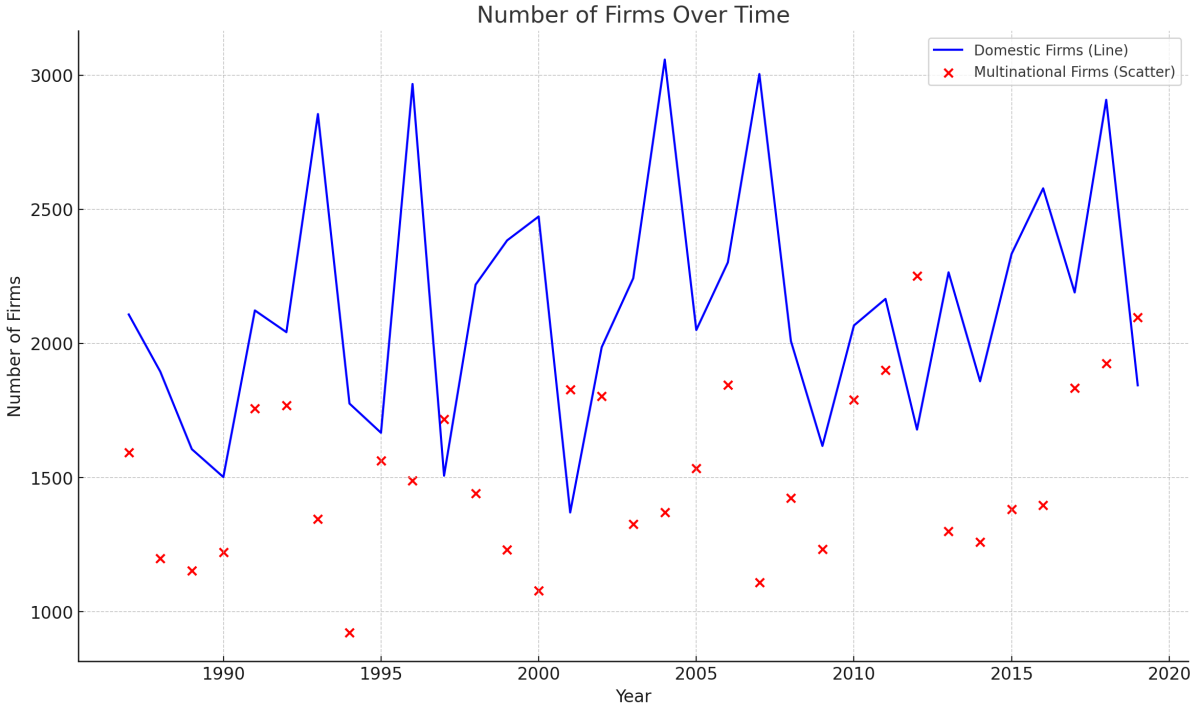

Figure 1: The frequency of both local and foreign businesses throughout time. The chart shows the total number of domestic and international businesses from January 1987 to December 2019—the start of our sample period (Dallin M. Alldredge, 2023)

The study also delves into the number of insiders trading during purchase and sale months, finding that the number of insiders trading in multinational firm purchase months is like that in domestic firm purchase months. However, it is observed that insiders at multinational firms tend to purchase and sell more shares compared to their domestic counterparts during these months. This observation points to potentially greater market activities and opportunities in multinational firms. The temporal aspect of the data is illustrated through figures showing the frequency of multinational and domestic firms over the sample period, and the proportion of insider trade months accounted for by multinational firms. Despite a decline in the number of unique firms around the 2001 downturn, the number of multinational firms consistently rises, increasing by about 50% from the beginning to the end of the sample period. In contrast, domestic firms exhibit a mostly declining trend, with the final year of the sample recording the lowest number of domestic firms, more than 50% fewer than at the beginning of the period. This trend underscores the growing importance of multinational firms in the context of insider trading (Usman Ali, 2017).

The study also presents the proportion of all firms and insider trade months represented by multinational firms, demonstrating a significant rise throughout the sample period. By 2019,

70% of all firms in the sample are multinational, and 73% of all insider trade months are accounted for by multinational firms. These figures highlight the increasing dominance of multinational firms in insider trading activities over time. Industries are another critical aspect of the analysis. The study includes firms across the Fama and French 12 industry classifications, with multinational firms being more concentrated in business equipment and manufacturing industries, while domestic firms are relatively concentrated in finance and wholesale/retail shopping industries. This industry representation necessitates a careful approach in subsequent analyses, prompting the use of propensity score matching by industry. This matching process, based on firm characteristics such as market capitalization, book-to-market ratio, and cumulative past 3-month raw return, aims to create a more comparable sample of multinational and domestic firms (Dallin M. Alldredge, 2015).

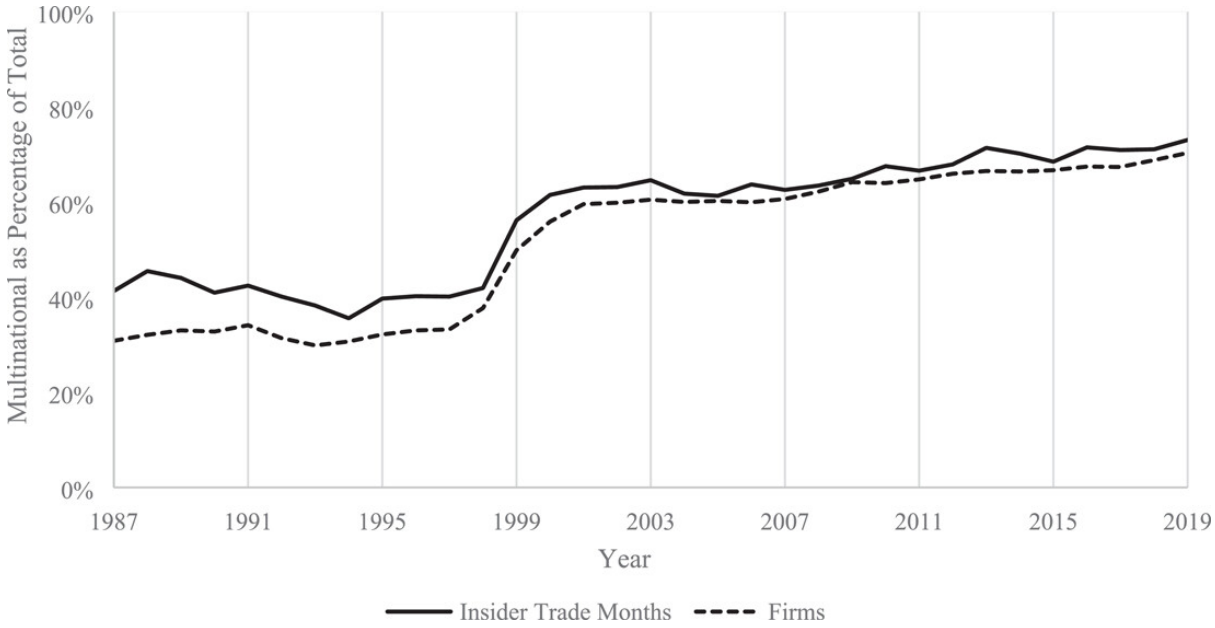

Figure 2: Proportion of international companies and months of insider trading. The graph shows the percentage of all firms and all insider trade months that are represented by global companies over the course of our sample, which runs from January 1987 to December 2019 (Dallin M. Alldredge, 2023).

The propensity scores matching employs a nearest neighbor one-to-one match with replacement, using a caliper of 2% of the standard deviation of the propensity score to filter matches. This rigorous matching process reduces the number of observations by about half but ensures that the differences between the matched groups are minimized. For instance, while the size of multinational firms is nearly double that of domestic firms before matching, the matched sample brings the size difference within 25%. Similarly, the book-to-market ratio difference is reduced to within 0.006, and past returns are twice as close. These adjustments

are critical for ensuring the robustness of the subsequent analyses. The empirical results begin with a univariate comparison of abnormal returns following trades at multinational and domestic firms. Abnormal returns are calculated as the 1-month raw stock return less the NYSE/AMEX/Nasdaq value-weighted market return. The analysis reveals that the abnormal return following multinational firm purchase months is 1.88%, compared to 1.38% for domestic firms, with a significant difference of 0.50% ($p = 0.008$). This comparison is further refined by examining top-level insiders (CEOs, Presidents, Chairmen) separately from other insiders. Top-level insiders at multinational firms generate abnormal returns of 2.57% following purchase months, compared to 1.61% at domestic firms, resulting in a significant difference of 0.96% ($p = 0.024$). For other insiders, the difference is 0.47% ($p = 0.018$), indicating that the information advantage is more pronounced for top-level insiders.

The study also finds that the dollar value of insider trading profits is 67% larger for multinational firms compared to domestic firms in the month following insider purchase months. This suggests that top-level insiders not only generate larger abnormal returns but also that these returns translate into substantial financial gains. However, the analysis of insider sale months reveals no significant differences in abnormal returns between multinational and domestic firms, indicating that the informational advantage is more pronounced in purchase transactions. The analysis is extended to multinational firms with varying levels of foreign sales. Firms with foreign sales accounting for greater than 25% of total sales are considered high foreign sales firms, while others are low foreign sales firms. The study finds that high foreign sales firms have larger returns during purchase months, although the differences are not statistically significant. However, for sale months, the return differences between high foreign sales firms and low foreign sales firms are statistically significant, especially for other (non-top-level) insiders. To formalize these findings, the study employs an Ordinary-Least-Squares (OLS) fixed-effect regression framework, controlling for other determinants of stock returns such as firm size, book-to-market ratio, market return, and past returns. The dependent variable is the excess 1-month return following the insider trading month. The key variables of interest are Multinational (an indicator for firms with foreign sales), High Foreign Sales (an indicator for multinational firms with foreign sales accounting for at least 25% of total sales), and Percent Foreign (the ratio of foreign sales to total sales). The regression results confirm the univariate findings, showing that abnormal returns following insider purchase months at multinational firms are significantly higher than those at domestic firms. The effect is particularly pronounced for firms with high foreign sales.

The study also analyzes insider returns by the insider's role within the firm. The results indicate that top-level insiders at multinational firms generate significantly higher abnormal returns following purchase months compared to their counterparts at domestic firms. This suggests that top-level insiders have greater access to value-relevant information, which they leverage to achieve higher trading returns. In contrast, no significant differences are observed for insider sales. The complexity of processing information about international sales is explored through the lens of linguistic and cultural differences. Following Huang (2015), the study categorizes international sales into three groups: English-speaking countries, non-English-speaking European countries, and Asian countries. The regression analysis reveals that insider trades at multinational firms with sales to Asian countries have higher return predictability compared to those with sales to English-speaking or non-English-speaking European countries. This suggests that linguistic and cultural differences pose greater challenges for outside investors, providing insiders with a trading advantage.

Overall, the study fills several gaps in the literature on insider trading. It highlights the differences in insider trading behaviors between multinational and domestic firms, emphasizing the greater information advantages and trading returns of top-level insiders at multinational firms. The study also underscores the impact of foreign sales on trading returns, with higher proportions of foreign sales enhancing insider trading profitability. The nuanced analysis of linguistic and cultural differences provides additional insights into how the complexity of international markets influences insider trading advantages. By employing robust methodologies such as propensity score matching and multivariate analysis, the study offers a comprehensive and reliable examination of insider trading dynamics in multinational and domestic firms. These findings contribute to a deeper understanding of the factors driving insider trading profitability and underscore the importance of considering firm-specific characteristics and market complexities in analyzing insider trading activities.

CHAPTER -4 CASE STUDIES (Ai Manipulation in Real World)

Case 1: Raj Rajaratnam and Galleon Group

One of the biggest insider trading cases in US history began when Raj Rajaratnam was detained and accused of insider trading in 2009. Rajaratnam traded stocks and made large profits totalling more than $60 million by using insider information. Rajaratnam received confidential information from several people, including insiders, investor relations firm staff, and company officials. This material was related to publicly traded firms' earnings releases and mergers and acquisitions. Using this private information, he made transactions that brought the Galleon Group large profits. For example, he was informed before it was made public that Advanced Micro Devices (AMD) was purchasing ATI Technologies, and he made trades based on this confidential information.

Former director of Procter & Gamble and Goldman Sachs, Gupta gave Rajaratnam access to private board meeting notes that included information regarding Warren Buffett's $5 billion investment in Goldman Sachs during the 2008 financial crisis. Senior partner at McKinsey & Company, Kumar provided Rajaratnam with confidential knowledge about multiple businesses. The case brought to light important problems with market integrity and the requirement for strict laws to stop insider trading. Insider trading was detected and discouraged by the U.S. Securities and Exchange Commission (SEC) and other regulatory agencies through heightened inspection and stronger enforcement. Because of the case's widespread media coverage, more people are now aware of insider trading's immoral and unlawful origins as well as its negative effects (Lattman & Ahmed, 2011).

Case 2: Knight Capital Group Trading Incident

Knight Capital Group was a major player in market-making and high-frequency trading (HFT). On August 1, 2012, Knight Capital experienced a massive trading glitch that led to significant financial losses and market disruption. One of the main participants in market-making and high-frequency trading (HFT) was Knight Capital Group. A huge trading error at Knight Capital on August 1, 2012, caused considerable losses in value and upset the market. The introduction of new trading software was the cause of the incident. To manage trading on the

New York Stock Exchange (NYSE), Knight Capital has deployed a new code. However, the system started to generate incorrect buy and sell orders because of a fundamental flaw in the software deployment. The rogue algorithm forced Knight Capital to make millions of trades at a breakneck speed over the course of 45 minutes, buying and selling huge volumes of equities without following any logical plan. The market was disturbed by this unusual trading behaviour, which caused 148 different equities' prices to fluctuate dramatically.

The incorrect trades caused Knight Capital to lose an incredible $440 million in just 45 minutes. The company's financial health was negatively harmed by this enormous loss. Significant market disruption resulted from the incident, which also damaged investor confidence in algorithmic trading systems and impacted the pricing of many equities. The incident drew regulatory attention and brought attention to the necessity for stronger risk control and supervision procedures for algorithmic trading and high-frequency trading.

Knight Capital needed an emergency capital infusion from a group of investors to stay out of bankruptcy. Finally, another high-frequency trading company, Getco LLC, bought the company, and it ultimately became a part of KCG Holdings. Because of the occurrence, regulators are now paying more attention to the dangers of algorithmic and high-frequency trading. The significance of pre-deployment testing, system protections, and real-time monitoring of trading algorithms was underscored by regulatory agencies such as the SEC to avert such disasters in the future. The financial industry learnt a valuable lesson from the Knight Capital fiasco, which highlighted the possible hazards associated with algorithmic trading. It made a lot of companies reevaluate their testing processes, trading system robustness, and risk management policies (Henricodolfing, 2019).

Case 3: Use of AI in Financial Fraud to Support Illicit Activities

Concern over hackers' use of AI and machine learning to increase the complexity and scope of their operations has grown in recent years. With the use of these technologies, thieves may streamline and automate many parts of their fraudulent operations, making them more profitable and difficult to uncover. Even though clear connections to terrorism are frequently kept a secret from the public, money from these kinds of operations can and often does go to terrorist groups.

Cybercriminals use email as part of BEC scams to trick businesses into sending huge amounts of money. These systems have historically relied on straightforward social engineering techniques, but AI has greatly boosted their efficacy. An advanced deepfake audio scam

victimized a UK-based energy company in 2019. The CEO of the UK-based company got a call from the CEO of the parent company in Germany, who he took to be his supervisor. The caller told him to transfer €220,000 to a Hungarian supplier using deepfake audio produced by AI. The UK CEO obeyed the request since the voice was so convincing, only to discover afterwards that it was a ruse.

Cybercriminals create extremely personalized and convincing phishing messages by using artificial intelligence (AI) to analyze massive volumes of data from emails, social media, and other sources. The purpose of these messages is to trick staff members into disclosing private information or accepting unauthorized transactions. Deepfakes of audio and video that are incredibly realistic are made with AI. In the instance of the energy company, machine learning algorithms trained on publicly accessible recordings of the CEO's speech were used to create deepfake audio that accurately resembled the CEO's voice. €220,000 was lost by the UK energy firm in the fraudulent transaction. Other businesses around the world have lost millions of dollars because of similar frauds. Although there isn't a clear connection between this specific instance and terrorism, it is well known that money earned through financial fraud can be used to finance terrorist attacks. To make money, criminal networks frequently commit a variety of financial crimes, some of which may be utilized to fund terrorism. To identify and stop this kind of fraud, businesses are using AI-driven security solutions more and more. These systems recognize odd patterns and behaviours that might point to fraud using machine learning. In their increased efforts to combat financial fraud and cybercrime, governments and regulatory organizations are utilizing artificial intelligence (AI) and other cutting-edge technology in their enforcement and investigative operations. Companies are spending money on training initiatives to teach staff members how to spot suspicious activity and how to protect themselves from AI-driven fraud.

Case 4- Hindustan Unilever Ltd. vs. Securities and Exchange Board of India

The lawsuit concerns HUL's parent firm, Unilever Plc, purchasing shares of a group subsidiary, Brooke Bond Lipton India Ltd. (BBLIL), during HUL and BBLIL's merger. Whether this purchase qualified as insider trading was the main question in the case. HUL was accused by SEBI of breaking insider trading laws when BBLIL and HUL merged. According to SEBI, Unilever Plc obtained unpublished price-sensitive information (UPSI) about the transaction and utilized it to purchase BBLIL shares before the information was made public. HUL defended itself by saying that Unilever Plc's purchase of BBLIL shares was a strategic move

that was carried out in full transparency. They argued that since the information was not secret at the time of the share purchase, there had been no insider trading. After alleging that HUL had violated insider trading regulations, SEBI launched an inquiry and issued orders against the company, imposing fines. The order made clear that HUL and its parent business had profited financially from their access to vital non-public information. In front of the Securities Appellate Tribunal (SAT), HUL contested SEBI's ruling. The business maintained that insider information was not misused and that the acquisition was a legal transaction. After reviewing the case, the Securities Appellate Tribunal (SAT) concluded that there was not enough evidence provided by SEBI to establish that the transactions were the result of insider knowledge. HUL was declared the winner by the SAT, which concluded that insider trading laws had not been broken. The SEBI's order and the fines levied against HUL were overturned by the SAT.SEBI filed an appeal with the Indian Supreme Court, citing dissatisfaction with the SAT's decision. The Supreme Court maintained the SAT's ruling, ruling that SEBI had not proven the disputed transactions were founded on insider knowledge (Mehta, 2017).

This case is important because it illustrates the difficulties in enforcing laws against insider trading and the high standard of proof that regulators such as SEBI demand. It emphasizes how crucial it is to have solid proof when accusing someone of insider trading, and how regulators must demonstrate that the accused had access to confidential, price-sensitive information and had profited from it. The case also serves as an example of the checks and balances in the Indian legal system, which allow courts and appellate bodies to review and contest regulatory orders (Saraswat, 2023).

CHAPTER -5 Methodology Analysis

The methodology employed in the insider trading study is both comprehensive and rigorous, integrating multiple data sources and sophisticated analytical techniques to explore insider trading behaviors in multinational and domestic firms over an extensive period from 1987 to 2019. The study utilizes insider trade data from Thomson Reuters insider Form 4 filings, stock return data from the Center for Research in Security Prices (CRSP), and financial statement data from Compustat. This combination of datasets ensures a robust foundation for the analysis, allowing for detailed examination of insider trading patterns and their financial implications. The selection criteria for the sample are methodically crafted to enhance the validity of the findings. By excluding firms with a market capitalization below $25 million, the study mitigates the influence of extreme outliers that could skew the results. Additionally, the winsorization of control variables at the 99% and 1% levels further refines the dataset, reducing the impact of anomalous data points and ensuring a more reliable analysis. A key feature of the methodology is the definition of insider trade months at the firm-month level, distinguishing between purchase months and sale months. This clear dichotomy allows for precise identification and classification of insider trading activities, facilitating targeted analysis. The differentiation of firms into multinational and domestic categories based on foreign sales presence is crucial for the study's focus on comparative insider trading behaviors. This classification enables the investigation of how international operations influence insider trading dynamics.

Excluding routine insiders using the methodology of Cohen et al. (2012) is a significant methodological decision, aiming to isolate trades driven by unique, non-routine information that insiders may possess. This exclusion enhances the focus on informed insider trading, providing a clearer picture of the impact of genuine informational advantages on trading outcomes. The summary statistics of the sample reveal a substantial and well-distributed dataset, with a significant portion of the data representing multinational firms. This distribution is vital for ensuring the robustness of comparative analyses between multinational and domestic firms. The study's approach to examining the number of insiders trading and the volume of shares traded during purchase and sale months provides valuable insights into the intensity and scale of insider trading activities. The observation that insiders at multinational firms tend to trade more shares compared to their domestic counterparts suggests greater

market activities and opportunities within multinational firms. This aspect of the analysis is essential for understanding the operational and strategic differences between multinational and domestic firms. The temporal dimension of the dataset is well-illustrated through figures showing the frequency of multinational and domestic firms over the sample period. The notable trend of increasing multinational firms and declining domestic firms underscores the growing importance of multinational operations in the context of insider trading. This trend provides a dynamic backdrop for analyzing the evolving landscape of insider trading activities.

The industry analysis, incorporating firms across the Fama and French 12 industry classifications, adds another layer of depth to the study. By noting the concentration of multinational firms in business equipment and manufacturing industries and domestic firms in finance and wholesale/retail shopping industries, the study acknowledges the need for careful industry-specific considerations. The use of propensity score matching by industry, based on firm characteristics such as market capitalization, book-to-market ratio, and cumulative past 3-month raw return, is a robust methodological step to create a comparable sample of multinational and domestic firms. This matching process, employing a nearest neighbor one-to-one match with replacement and a caliper of 2% of the standard deviation of the propensity score, is rigorous and ensures that the differences between the matched groups are minimized, thereby enhancing the validity of subsequent analyses. The empirical results begin with a univariate comparison of abnormal returns following trades at multinational and domestic firms, using abnormal returns calculated as the 1-month raw stock return less the NYSE/AMEX/Nasdaq value-weighted market return. This approach to calculating abnormal returns is standard in financial research and provides a clear measure of the performance impact of insider trading activities. The finding that abnormal returns following multinational firm purchase months are significantly higher than those for domestic firms, particularly for top-level insiders, underscores the informational advantages leveraged by insiders in multinational firms. This distinction between top-level insiders and other insiders further refines the analysis, highlighting the varying degrees of informational advantages within firms. The analysis of insider trading profits, indicating that the dollar value of insider trading profits is substantially larger for multinational firms, aligns with the abnormal return findings and reinforces the financial significance of insider trading activities. This financial quantification is crucial for understanding the real-world implications of the study's findings.

Extending the analysis to multinational firms with varying levels of foreign sales provides additional granularity. The differentiation between high foreign sales firms and low foreign sales firms allows for an examination of how the extent of international operations influences insider trading profitability. The use of an Ordinary-Least-Squares (OLS) fixed-effect regression framework to control for other determinants of stock returns, such as firm size, book-to-market ratio, market return, and past returns, adds robustness to the analysis. This multivariate approach ensures that the findings are not confounded by other factors, providing a clearer attribution of abnormal returns to insider trading activities. The study's exploration of linguistic and cultural differences in processing information about international sales, following Huang (2015), is an innovative methodological aspect. By categorizing international sales into English-speaking countries, non-English-speaking European countries, and Asian countries, the study provides nuanced insights into how these differences influence insider trading advantages. The regression analysis showing higher return predictability for insider trades at multinational firms with sales to Asian countries highlights the complexities of international markets and the trading advantages they afford insiders.

Detailed Analysis of Key Methodological Aspects

Data Sources and Time Period:

The study's utilization of multiple authoritative data sources—Thomson Reuters insider Form 4 filings, CRSP stock return data, and Compustat financial statements—provides a robust and reliable dataset. Covering an extensive period from 1987 to 2019, the study can capture long-term trends and patterns in insider trading, allowing for a more comprehensive understanding of how insider trading behaviors have evolved over time. This extensive timeframe also helps mitigate the impact of short-term market anomalies, providing a more stable basis for analysis.

Sample Selection and Refinement:

By excluding firms with a market capitalization below $25 million, the study effectively reduces the influence of extreme outliers that could distort the findings. This exclusion criterion ensures that the sample is more representative of larger, more established firms where insider trading activities are likely to be more significant and observable. Additionally, winsorizing

control variables at the 99% and 1% levels further refines the dataset by mitigating the impact of outliers, ensuring that the analysis is not unduly influenced by extreme values.

Definition of Insider Trade Months:

The clear distinction between purchase months and sale months at the firm-month level allows for a precise classification of insider trading activities. This approach facilitates a targeted analysis of the different types of insider trading, enabling the study to explore how insider purchases and sales impact stock returns differently. The meticulous structuring of insider trade months around net purchasing and net selling activities provides a solid foundation for subsequent analyses.

Classification of Firms:

The classification of firms into multinational and domestic categories based on the presence of foreign sales is a crucial methodological step. This classification enables the study to investigate how international operations influence insider trading behaviors, providing insights into the unique challenges and opportunities faced by multinational firms. The clear distinction between multinational and domestic firms also allows for a more focused analysis of the differences in insider trading dynamics between these two groups.

Exclusion of Routine Insiders:

Excluding routine insiders using the methodology of Cohen et al. (2012) is a significant methodological decision aimed at isolating trades driven by unique, non-routine information. This exclusion enhances the focus on informed insider trading, ensuring that the analysis captures trades based on genuine informational advantages rather than routine trading activities. This approach is pivotal for providing a clearer picture of the impact of insider information on trading outcomes.

Summary Statistics:

The summary statistics of the sample reveal a substantial and well-distributed dataset, with a significant portion of the data representing multinational firms. This distribution is vital for

ensuring the robustness of comparative analyses between multinational and domestic firms. The detailed summary statistics provide valuable insights into the characteristics of the sample, such as firm size, book-to-market ratios, and return volatility, which are crucial for understanding the context of the analysis.

Number of Insiders Trading and Volume of Shares Traded:

The study's approach to examining the number of insiders trading and the volume of shares traded during purchase and sale months provides valuable insights into the intensity and scale of insider trading activities. The observation that insiders at multinational firms tend to trade more shares compared to their domestic counterparts suggests greater market activities and opportunities within multinational firms. This aspect of the analysis is essential for understanding the operational and strategic differences between multinational and domestic firms.

Temporal Dimension:

The temporal dimension of the dataset is well-illustrated through figures showing the frequency of multinational and domestic firms over the sample period. The notable trend of increasing multinational firms and declining domestic firms underscores the growing importance of multinational operations in the context of insider trading. This trend provides a dynamic backdrop for analyzing the evolving landscape of insider trading activities.

Industry Analysis:

The industry analysis, incorporating firms across the Fama and French 12 industry classifications, adds another layer of depth to the study. By noting the concentration of multinational firms in business equipment and manufacturing industries and domestic firms in finance and wholesale/retail shopping industries, the study acknowledges the need for careful industry-specific considerations. The use of propensity score matching by industry, based on firm characteristics such as market capitalization, book-to-market ratio, and cumulative past 3-month raw return, is a robust methodological step to create a comparable sample of multinational and domestic firms. This matching process, employing a nearest neighbor one-to-one match with replacement and a caliper of 2% of the standard deviation of the propensity

score, is rigorous and ensures that the differences between the matched groups are minimized, thereby enhancing the validity of subsequent analyses.

Empirical Results:

The empirical results begin with a univariate comparison of abnormal returns following trades at multinational and domestic firms, using abnormal returns calculated as the 1-month raw stock return less the NYSE/AMEX/Nasdaq value-weighted market return. This approach to calculating abnormal returns is standard in financial research and provides a clear measure of the performance impact of insider trading activities. The finding that abnormal returns following multinational firm purchase months are significantly higher than those for domestic firms, particularly for top-level insiders, underscores the informational advantages leveraged by insiders in multinational firms. This distinction between top-level insiders and other insiders further refines the analysis, highlighting the varying degrees of informational advantages within firms.

Analysis of Insider Trading Profits:

The analysis of insider trading profits, indicating that the dollar value of insider trading profits is substantially larger for multinational firms, aligns with the abnormal return findings and reinforces the financial significance of insider trading activities. This financial quantification is crucial for understanding the real-world implications of the study's findings. The larger profits generated by insiders at multinational firms suggest that these firms provide greater opportunities for profitable insider trading, likely due to their more complex and expansive operations.

Extension to Firms with Varying Levels of Foreign Sales:

Extending the analysis to multinational firms with varying levels of foreign sales provides additional granularity. The differentiation between high foreign sales firms and low foreign sales firms allows for an examination of how the extent of international operations influences insider trading profitability. The use of an Ordinary-Least-Squares (OLS) fixed-effect regression framework to control for other determinants of stock returns, such as firm size, book-to-market ratio, market return, and past returns, adds robustness to the analysis. This

multivariate approach ensures that the findings are not confounded by other factors, providing a clearer attribution of abnormal returns to insider trading activities.

Linguistic and Cultural Differences:

The study's exploration of linguistic and cultural differences in processing information about international sales, following Huang (2015), is an innovative methodological aspect. By categorizing international sales into English-speaking countries, non-English-speaking European countries, and Asian countries, the study provides nuanced insights into how these differences influence insider trading advantages. The regression analysis showing higher return predictability for insider trades at multinational firms with sales to Asian countries highlights the complexities of international markets and the trading advantages they afford insiders.

Overall, the methodology of this insider trading study is robust, well-structured, and comprehensive. The combination of detailed data collection, rigorous sample refinement, sophisticated analytical techniques, and innovative explorations of contextual factors provides a thorough examination of insider trading dynamics. By employing robust methodologies such as propensity score matching and multivariate analysis, the study offers reliable and insightful findings that contribute significantly to the understanding of insider trading behaviors in multinational and domestic firms. The study’s methodological rigor ensures the validity and reliability of its conclusions, making it a valuable addition to the literature on insider trading.

CHAPTER – 6 CONCLUSION

The study finds that insiders at multinational firms engage in larger trades that are followed by larger abnormal returns compared to those at domestic firms. This suggests that insiders have a strategic edge in their trading operations and that they have better opportunities to benefit from their trades in international environments. According to the study, insiders often buy cheap, value stocks with poor historical performance. This conduct is indicative of a methodical strategy wherein insiders use their understanding of the firm's worth and inefficiencies in the market to optimize their trading gains. A crucial conclusion is that the timing of trades, especially before earnings announcements, significantly enhances the profitability of insider trading at multinational firms. This timing advantage allows insiders to capitalize on information that is not yet available to the broader market, further increasing their trading success. According to the study, multinational companies that have overseas sales in areas that are culturally and linguistically different from the United States tend to make the most money from insider trading. This implies that insiders are successfully taking advantage of intricate foreign information that could be challenging for other market players to comprehend, resulting in increased trading benefits. Important problems concerning market efficiency, and the moral implications of insider trading are brought up by the findings. Insiders' potential to profit by having access to information that is difficult to comprehend raises the possibility that markets don't always function efficiently, especially when it comes to multinational corporations.

According to the study, insiders at multinational corporations can make lucrative trades by taking advantage of intricate foreign knowledge. This implies that to more successfully monitor insider trading activity, regulators might need to bolster their monitoring procedures. Stricter reporting guidelines and increased transparency could help reduce the possibility of insider information being abused. Investors need to understand the particulars of insider trading in global corporations. The results imply that insiders make bigger and more lucrative deals, which might be a sign of possible market mispricing. To protect themselves against the negative effects of insider trading, investors may want to think about implementing more cautious investing tactics, especially when assessing the stocks of international corporations. To lower the risks related to insider trading, multinational corporations might need to review their corporate governance procedures. To prevent insiders from abusing their positions for personal benefit, it can be helpful to establish stricter internal controls and ethical standards. This could entail educating staff members on the moral ramifications of insider trading and the

value of preserving financial reporting openness. The results of the study cast doubt on market efficiency, especially considering multinational corporations. Insiders' capacity to profit from their access to hard-to-process information raises the possibility that markets don't always function effectively. When assessing the stock performance of multinational corporations, analysts and investors should take the possibility of information asymmetry into account because it may cause mispricing in the market. The fact that insider trading profits are largest in companies with overseas sales in culturally diverse areas suggests that multinational corporations should operate with cultural sensitivity. Firms can better negotiate the challenges of operating in varied locations by having a better understanding of local market dynamics and the possibility of insider trading, which will eventually improve risk management and compliance standards.

This study creates opportunities for more investigation into the nuances of insider trading in other settings in the future. Based on the results, the following recommendations for future research have been made. To gain more insight into the variations in insider trading practices, comparative analyses between domestic and multinational corporations may be carried out in subsequent study. This could entail examining the elements that influence insider trade profitability in various business models, offering a more thorough understanding of the dynamics of insider trading in a range of market configurations. Understanding how insider trading behaviors change over time, particularly in reaction to shifting market conditions, regulatory frameworks, and corporate governance procedures, may be aided by conducting longitudinal studies. This strategy might shed light on how insider trading affects market efficiency and company profitability over the long run. Subsequent investigations may investigate how regulatory modifications affect insider trading in global corporations. Researchers might evaluate the effects of major regulatory improvements on insider trading profitability and market behavior by comparing periods before to and following the reforms. A deeper comprehension of the motivations for insider trading may be achieved by looking into the psychological and behavioral characteristics of insiders in multinational corporations. Investigating how cognitive biases and decision-making processes affect trading behaviors, particularly when dealing with sophisticated foreign information, may fall under this category. Future studies should examine how technological developments affect insider trading behaviors considering the growing importance of data analytics and technology. This can entail

researching how big data and machine learning might be used to spot trading trends and anticipate insider trading activity in large corporations.

Whistleblowers play a crucial role in exposing corruption, fraud, and misconduct, but their protection is often precarious. Legal safeguards, such as whistleblower protection laws, are designed to shield them from retaliation, including job loss or harassment. However, these protections can be limited by loopholes, weak enforcement, and organizational culture that discourages dissent. Effective protection requires robust legal frameworks, strong institutional support, and a societal commitment to transparency and accountability. Without these, whistleblowers risk significant personal and professional harm, which can deter others from coming forward, ultimately undermining efforts to hold powerful entities accountable. The governments of every nation should enact standard laws to safeguard whistleblowers from harm to end this conflict and legal loopholes.

However, several adjustments are necessary to improve AI-powered trading and anti-money laundering (AML) systems for preventing fraud. First, by incorporating real-time data analytics, trade patterns and transactions may be continuously monitored, enabling the quick identification of anomalies. To enhance the identification of intricate fraud schemes, a variety of datasets should be used to train complex machine learning models. Incorporating explainable AI also guarantees openness, allowing regulators to comprehend and have confidence in AI-driven choices. Standardizing AI implementation through cooperation between regulators and financial institutions can guarantee compliance and uniformity. Updating often and implementing strong cybersecurity defenses are essential to guard against changing fraud strategies.

APPENDIX

Table 1: Summary Statistics

Panel	Metric	Mean	Median	Std Dev	5th Percentile	95th Percentile
A	**Multinational firm purchase months (3928 firms; 31,721 firm-month observations)**					
	B/M	0.55	0.46	0.39	0.10	1.30
	Size	3961	596	10,442	44	20,565
	Return Vol	0.033	0.027	0.023	0.011	0.077
	Traders per firm-month	1.48	1.00	1.43	1.00	3.00
	Shares traded per insider	30,209	3000	226,305	100	79,600
B	**Multinational firm sale months (4273 firms; 82,412 firm-month observations)**					
	B/M	0.42	0.35	0.31	0.07	0.98
	Size	5886	1285	12,264	79	32,186
	Return Vol	0.025	0.021	0.017	0.009	0.056
	Traders per firm-month	1.57	1.00	1.14	1.00	4.00
	Shares traded per insider	55,423	9360	384,194	500	180,000
C	**Domestic firm purchase months (5014 firms; 35,527 firm-month observations)**					
	B/M	0.61	0.53	0.41	0.11	1.42
	Size	1105	209	3818	33	4233
	Return Vol	0.034	0.028	0.024	0.010	0.078
	Traders per firm-month	1.46	1.00	1.09	1.00	3.00
	Shares traded per insider	20,965	2100	222,893	100	57,517
D	**Domestic firm sale months (5147 firms; 57,894 firm-month observations)**					
	B/M	0.47	0.40	0.34	0.08	1.12
	Size	1789	369	5464	39	7053
	Return Vol	0.029	0.025	0.019	0.010	0.060
	Traders per firm-month	1.53	1.00	1.08	1.00	4.00
	Shares traded per insider	48,243	9921	291,888	500	172,125

Table 2: Multinational and domestic sample distribution of industry.

Industry	Multinational Number of firms	Multinational Percentage (%)	Domestic Number of firms	Domestic Percentage (%)	Difference Percentage (%)
Business equipment	1326	28.30	914	15.43	12.87
Manufacturing	621	13.25	492	8.31	4.94
Chemicals	136	2.90	72	1.22	1.68
Consumer durables	154	3.29	115	1.94	1.35
Consumer nondurables	255	5.44	293	4.95	0.49
Energy	152	3.24	266	4.49	-1.25
Other	831	17.73	1127	19.03	-1.30
Telephone and television	114	2.43	260	4.39	-1.96
Utilities	63	1.34	202	3.41	-2.07
Healthcare	415	8.86	691	11.67	-2.81
Finance	248	5.29	615	10.39	-5.10
Shops—wholesale and retail	371	7.92	875	14.78	-6.86
Total	**4686**	**100**	**5922**	**100**	—

Table 3. Abnormal returns following insider trade months at multinational firms.

Panel A: Multinational and Domestic Firm Months

Metric	All Insider Purchase Months (1)	Top-Level Insider Purchase Months (2)	Other Insider Purchase Months (3)	All Insider Sale Months (4)	Top-Level Insider Sale Months (5)	Other Insider Sale Months (6)
Multinational firms						
Abnormal returns (%)	1.8	2.57	1.93	-0.10	-0.21	-0.11
Observations	22,509	5840	20,169	53,752	11,670	48,693
Domestic firms						
Abnormal returns (%)	1.38	1.61	1.46	-0.10	-0.06	-0.14
Observations	12,168	5840	10,992	23,925	5439	21,604
Multinational–domestic difference (%)	0.50***	0.96**	0.47**	0.00	-0.14	-0.03
p Value	(0.008)	(0.024)	(0.018)	(0.995)	(0.566)	(0.789)

Panel B: Multinational Firm Months

Metric	All Insider Purchase Months (1)	Top-Level Insider Purchase Months (2)	Other Insider Purchase Months (3)	All Insider Sale Months (4)	Top-Level Insider Sale Months (5)	Other Insider Sale Months (6)
High foreign sales firms						
Abnormal returns (%)	1.91	2.59	1.96	-0.22	-0.25	-0.25
Observations	10,960	2892	9837	28,209	5916	25,660
Low foreign sales firms						
Abnormal returns (%)	1.84	2.54	1.90	0.03	-0.16	0.06
Observations	11,549	2948	10,332	25,543	5754	23,033
High foreign sales–low foreign sales difference (%)	0.07	0.05	0.06	-0.25**	-0.08	-0.31**
p Value	(0.762)	(0.920)	(0.805)	(0.031)	(0.761)	(0.011)

Table 4. Multivariate test of insider trading in multinational and domestic firms.

Panel A: Matched Sample

Metric	Purchase Months (1)	Purchase Months (2)	Purchase Months (3)	Sale Months (4)	Sale Months (5)	Sale Months (6)
Multinational	0.0047**			-0.0002		
	(0.013)			(0.877)		
High Foreign Sales		0.0032*			-0.0023**	
		(0.092)			(0.023)	
Percent Foreign			0.0079*			-0.0055**
			(0.057)			(0.010)
Log(Size)	-0.0038***	-0.0037***	-0.0037***	-0.0014***	-0.0013***	-0.0013***
	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(0.002)	(0.002)
B/M	0.0047	0.0049	0.0049	0.0008	0.0009	0.0010
	(0.126)	(0.110)	(0.111)	(0.651)	(0.618)	(0.603)
Past Ret	-0.0153***	-0.0153***	-0.0153***	0.0165***	0.0165***	0.0165***
	(0.004)	(0.004)	(0.004)	(<0.001)	(<0.001)	(<0.001)
Market	1.436***	1.436***	1.436***	1.378***	1.379***	1.379***
	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Constant	0.0152***	0.0165***	0.0162***	-0.0222***	-0.0222***	-0.0220***
	(0.009)	(0.004)	(0.005)	(<0.001)	(<0.001)	(<0.001)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	34,676	34,676	34,676	77,678	77,678	77,678
R^2	0.137	0.137	0.137	0.127	0.127	0.127

Panel B: Full Sample

Metric	Purchase Months (1)	Purchase Months (2)	Purchase Months (3)	Sale Months (4)	Sale Months (5)	Sale Months (6)
Multinational	0.0039***			0.0007		
	(0.002)			(0.368)		
High Foreign Sales		0.0031**			-0.0008	
		(0.032)			(0.280)	
Percent Foreign			0.0065**			-0.0018
			(0.029)			(0.223)
Log(Size)	-0.0034***	-0.0033***	-0.0033***	-0.0003	-0.0002	-0.0002
	(<0.001)	(<0.001)	(<0.001)	(0.231)	(0.468)	(0.482)
B/M	0.0044**	0.0044**	0.0044**	0.0030**	0.0030**	0.0030**
	(0.034)	(0.034)	(0.033)	(0.018)	(0.019)	(0.019)
Past Ret	-0.0185***	-0.0186***	-0.0185***	0.0170***	0.0170***	0.0170***
	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Market	1.334***	1.334***	1.334***	1.301***	1.301***	1.301***
	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)	(<0.001)
Constant	0.0070**	0.0074**	0.0073**	-0.0279***	-0.0280***	-0.0280***
	(0.048)	(0.038)	(0.041)	(<0.001)	(<0.001)	(<0.001)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	67,248	67,248	67,248	140,306	140,306	140,306
R^2	0.124	0.124	0.124	0.127	0.127	0.127

Table 5. Multivariate test of top-level insiders and other insiders.

Panel	Variable	Top-level insider purchases (1)	Other insider purchases (2)	Top-level insider sales (3)	Other insider sales (4)
Panel A: Matched sample	Multinational	0.0096** (0.026)	0.0043** (0.030)	−0.0010 (0.710)	0.0001 (0.923)
	Log(Size)	−0.0044*** (0.002)	−0.0041*** (<0.001)	−0.0030*** (0.003)	−0.0012*** (0.005)
	B/M	0.0060 (0.381)	0.0045 (0.170)	−0.0049 (0.238)	0.0005 (0.803)
	Past Ret	−0.0138 (0.254)	−0.0187*** (0.001)	0.0136** (0.047)	0.0140*** (<0.001)
	Market	1.508*** (<0.001)	1.442*** (<0.001)	1.458*** (<0.001)	1.380*** (<0.001)
	Constant	0.0234 (0.112)	0.0169*** (0.006)	−0.0091 (0.366)	−0.0227*** (<0.001)
	Year FE	Yes	Yes	Yes	Yes
	Observations	8756	31,161	17,109	70,297
	R^2	0.139	0.137	0.120	0.128
Panel B: Full sample	Multinational	0.0071** (0.018)	0.0040*** (0.003)	0.0014 (0.430)	0.0008 (0.308)
	Log(Size)	−0.0035*** (<0.001)	−0.0035*** (<0.001)	−0.0009 (0.123)	−0.0003 (0.263)
	B/M	0.0064 (0.138)	0.0053** (0.020)	−0.0020 (0.505)	0.0028** (0.037)
	Past Ret	−0.0194** (0.018)	−0.0203*** (<0.001)	0.0151*** (0.002)	0.0150*** (<0.001)
	Market	1.439*** (<0.001)	1.331*** (<0.001)	1.364*** (<0.001)	1.303*** (<0.001)
	Constant	0.0118 (0.187)	0.0073* (0.052)	−0.0216*** (0.001)	−0.0271*** (<0.001)
	Year FE	Yes	Yes	Yes	Yes
	Observations	15,887	60,866	29,537	127,672
	R^2	0.134	0.124	0.117	0.127

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

This table summarizes the results from the regressions in both the matched sample (Panel A) and the full sample (Panel B) for different types of insider purchases and sales. The coefficients are presented along with p-values in parentheses. The significance levels are indicated by stars, where *** denotes significance at the 1% level, ** at the 5% level, and * at the 10% level.

Table 6. Linguistic or cultural difference and insider trading in multinational firms.

Panel	Variable	(1)	(2)	(3)	(4)
Panel A: Matched sample of multinational firms	**Asia**	0.0112*** (0.004)		0.0103*** (0.009)	
	Europe		−0.0088 (0.299)		−0.0075 (0.382)
	English			−0.0046 (0.195)	−0.0027 (0.456)
	Log(Size)	−0.0042*** (<0.001)	−0.0043*** (<0.001)	−0.0042*** (<0.001)	−0.0042*** (<0.001)
	B/M	0.0062 (0.104)	0.0057 (0.130)	0.0060 (0.113)	0.0063* (0.097)
	Past Ret	−0.0225*** (0.001)	−0.0225*** (0.001)	−0.0225*** (0.001)	−0.0224*** (0.001)
	Market	1.486*** (<0.001)	1.485*** (<0.001)	1.486*** (<0.001)	1.486*** (<0.001)
	Constant	0.0191** (0.011)	0.0201*** (0.007)	0.0197*** (0.009)	0.0192** (0.011)
	Year FE	Yes	Yes	Yes	Yes
	Observations	22,508	22,508	22,508	22,508
	R^2	0.150	0.149	0.149	0.150
Panel B: Full sample of multinational firms	**Asia**	0.0070** (0.014)		0.0061** (0.037)	
	Europe		−0.0029 (0.687)		−0.0025 (0.732)
	English			−0.0051* (0.076)	−0.0038 (0.195)
	Log(Size)	−0.0036*** (<0.001)	−0.0036*** (<0.001)	−0.0036*** (<0.001)	−0.0036*** (<0.001)
	B/M	0.0049* (0.097)	0.0047 (0.114)	0.0049* (0.096)	0.0051* (0.086)
	Past Ret	−0.0233*** (<0.001)	−0.0233*** (<0.001)	−0.0233*** (<0.001)	−0.0233*** (<0.001)
	Market	1.454*** (<0.001)	1.454*** (<0.001)	1.455*** (<0.001)	1.455*** (<0.001)
	Constant	0.0169***	0.0172***	0.0169***	0.0169***

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